Iterative In Order Traversal

AlgoExpert · Very Hard · Binary Trees

The problem

- In-order traversal with O(1) extra space using parent pointers.

Intuition

- Parent pointers let us know where we came from, so we can walk without a stack.

Approach

- 1. Track previous and current
- 2. Coming from parent, go left if possible
- 3. Coming from left, visit then go right
- 4. Coming from right, go up

Java solution

```java
public class IterativeInOrderTraversal {
    public static void iterativeInOrderTraversal(BinaryTree tree, Consumer<BinaryTree> callback) {
        BinaryTree prev = null, cur = tree;
        while (cur != null) {
            BinaryTree next;
            if (prev == null || prev == cur.parent) {
                if (cur.left != null) next = cur.left;
                else { callback.accept(cur); next = cur.right != null ? cur.right : cur.parent; }
            } else if (prev == cur.left) {
                callback.accept(cur);
                next = cur.right != null ? cur.right : cur.parent;
            } else next = cur.parent;
            prev = cur;
            cur = next;
        }
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Parent pointers let us know where we came from, so we can walk without a stack.
- Complexity: O(n)